

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 1 : Session – 3 : SLO – 2 | Multiprogramming / Time-Sharing Word Search + Kernel Structures

■ I. Word Search — Words to Find

- MONITOR, MULTITASKING, PROCESSOR, VIEW, ROUND ROBIN, SHORTEST, SIMULTANEOUSLY, TIME SHARING
- All eight words are located in the grid (horizontal, vertical, and diagonal directions).
- Hidden message from the unused letters: "MULTIPROGRAMMING IMPROVES CPU UTILIZATION".

■ II. Kernel Structures — Reverse Word Search (place these words)

- EXOKERNEL — minimalist kernel that exposes hardware to applications, giving them direct control.
- HYBRID KERNEL — combines features of monolithic and microkernels; used in Windows and macOS.
- LAYERED — organized in levels; each layer builds on services provided by the one below it.
- MICROKERNEL — only essential functions like IPC and basic scheduling; everything else in user space.
- MODULAR — components can be loaded and unloaded at runtime; used in Linux systems (LKMs).
- MONOLITHIC — single large process running entirely in kernel mode with all OS services included.
- SIMPLE — basic structure with minimal functionality, used in teaching or early systems (e.g. MS-DOS).
- VIRTUAL MACHINE (HYPERVISOR) — abstracts physical hardware; runs multiple OSes on the same system.